

Solutions to JEE Advanced -11 | JEE 2022 | Paper-1

PHYSICS

1.(ABC)

Initially when sand starts falling but it doesn't hit the surface DE .

$$\text{Reading} = (m_0 + M)g - \lambda tg$$

At time $t_1 = \sqrt{\frac{2h}{g}}$ sand starts hitting the surface

$$\text{Reading just before } t = t_1 \text{ is } (m_0 + M)g - \lambda g \sqrt{\frac{2h}{g}} = (m_0 + M)g - \lambda \sqrt{2gh}$$

When sand starts hitting the surface.

$$\text{Reading} = (m_0 + M)g + \lambda \sqrt{2gh} - \lambda \sqrt{\frac{2h}{g}}g = (m_0 + M)g$$

After time $t_2 = \frac{m_0}{\lambda}$ upper reservoir will be empty.

Reading till all sand hits DE (for $t > t_2$) is $= (m_0 + M + \lambda(t - t_2))g$

And for time $t > t_3$ (All sand has reached lower reservoir)

$$\text{Reading} = (M + m_0)g$$

2.(BCD)

$$5 \times 10^{-5} = (2 \times 10^{-6}) (2a(0.1)^2 - 0)$$

$$a = 1.25 \times 10^3 \frac{V}{m^2}$$

Any plane parallel to $x - y$ plane $z = \text{constant}$

$$V = ax^2 + ay^2 + \text{const.}$$

$$ax^2 + ay^2 = V_0 - \text{const.} = K \text{ (a circle)}$$

Substituting a, v and $z = \sqrt{2}$

$$6250 = (1.25 \times 10^3)(x^2 + y^2 + 2 \times 2) \Rightarrow x^2 + y^2 = 1$$

3.(ACD)

$$(i) \quad h = \frac{2\sigma \cos \theta}{\rho g} = \frac{2 \times 0.07 \times \cos 0^\circ}{(0.5 \times 10^{-3}) \times 10^3 \times 10} = 2.8 \times 10^{-2} m = 2.8 \text{ cm}$$

(ii) The vertical height h of water column will remain the same $= 2.8 \text{ cm}$
Length of water column is

$$l = \frac{h}{\cos 60^\circ} = 2h = 5.6 \text{ cm}$$

But only 5 cm of tube is available

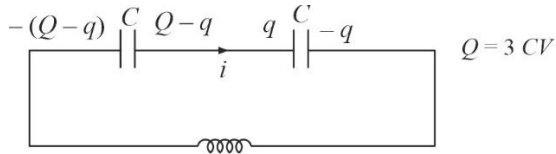
$$\therefore l = 5 \text{ cm}$$

When accelerating up (setup-2)

$$h = \frac{2\sigma \cos \theta}{r\rho \left(\frac{3g}{2}\right)} = 1.87 \text{ cm}; \quad l = \frac{h}{\cos 60^\circ} = 3.73 \text{ cm}$$

4.(ACD)

At $t = t$



$$i = \frac{dq}{dt}$$

$$\frac{-(Q_0 - q)}{C} + \frac{q}{C} + L \frac{di}{dt} = 0$$

$$\frac{2q - Q_0}{C} + \frac{L di}{dt} = 0$$

$$\frac{d^2 q}{dt^2} = -\frac{2}{LC} \left(q - \frac{Q_0}{2} \right)$$

$$\Rightarrow q = \frac{Q_0}{2} + A_0 \sin(\omega t + \phi)$$

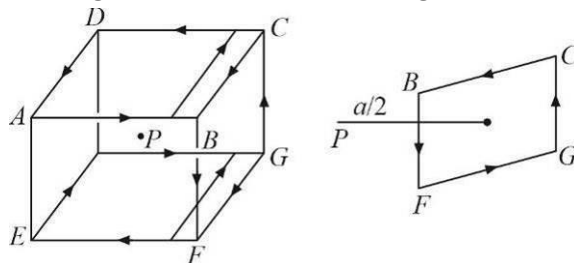
$$\omega = \sqrt{\frac{2}{LC}} \Rightarrow \frac{2\pi}{T} = \pi \sqrt{2LC}$$

$$\text{Initially } i = 0 \text{ and } q = CV \text{ so } \phi = \frac{\pi}{2} \text{ and } A_0 = \frac{CV}{2}$$

$$\text{So } q = \frac{3CV}{2} - \frac{CV}{2} \cos(\omega t)$$

$$\text{So } q_{\max} = \frac{3CV}{2} + \frac{CV}{2} = 2CV$$

5.(AD) The resultant figure can be redrawn as in figure.

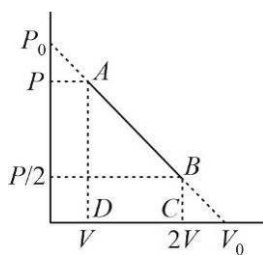


The field at P due to upper and lower squares will cancel out. Resultant field at P will due to the square $BCGF$ as shown in the figure. Solve to get, $B_P = \frac{2\mu_0 I}{\pi\sqrt{3}a}$

6.(ABC)

(A) Work done in process A to B is (see figure)

$$W_1 = \text{area of trapezium } ABCD = \left(P + \frac{P}{2} \right) V = \frac{3PV}{2} = \frac{3RT}{2} \text{ (for 1 mole)}$$



If the process A to B were isothermal, the work done would be

$$W_2 = RT \log_e \left(\frac{V_2}{V_1} \right) = RT \log_e (2) = 0.69RT$$

Thus, $W_1 > W_2$

So, choice (A) is correct

(B) Slope of line $= -\frac{P}{2V}$ = slope of adiabat at point (P', V')

$$\Rightarrow P' = \frac{3}{10} \frac{PV'}{V} \quad \dots\dots\dots (1)$$

Equation of line

$$P' = -\frac{PV'}{2V} + \frac{3P}{2} \quad \dots\dots\dots (2)$$

$$\Rightarrow V' = \frac{15V}{8}$$

(C) Since $V = \frac{RT}{P}$, Eq. (i) becomes

$$\frac{P}{P_0} + \frac{RT}{PV_0} = 1 \Rightarrow T = V_0 P - \frac{V_0 P^2}{RP_0} \quad \dots\dots\dots (ii)$$

Which represents a Parabola

So, choice is correct

(D) Let P_0 and V_0 be the intercepts on the P and V axes. The equation of straight line AB is

$$P = \frac{P_0}{V_0}(V - V_0)$$

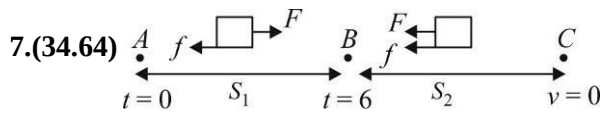
$$\Rightarrow \frac{P}{P_0} + \frac{V}{V_0} = 1 \quad \dots\dots\dots (i)$$

Since $P = \frac{RT}{V}$, Eq. (i) becomes

$$\frac{RT}{VP_0} + \frac{V}{V_0} = 1 \Rightarrow T = \frac{P_0 V}{R} - \frac{P_0 V^2}{RV_0}$$

Which represents a parabola on the $T - V$ graph

So temperature first increases reaches a maximum and then decreases.



From $A(t=0)$ to $B(t=6)$

$$a = \frac{F - f}{m} = 10^{-5} = 5 \text{ m/s}^2$$

$$s_1 = \frac{1}{2}at^2 = \frac{1}{2}(5)(36) = 90 \text{ m}$$

$$\text{Velocity at } B = at = 5(6) = 30 \text{ m/s}$$

After B , the direction of F is reversed.

As velocity is in forward direction, friction acts backwards till block comes to instantaneous rest at C .

From B to C

$$a = -\frac{(F + f)}{m} = -(10 + 5) = -15 \text{ m/s}^2$$

$$u = 30 \text{ m/s}$$

$$v^2 = u^2 + 2as_2$$

$$0 = (30)^2 + 2(-15)s_2 \Rightarrow s_2 = 30 \text{ m}$$

While returning from C to A , F acts towards left but friction acts towards right (as velocity is towards left)

$$s = -(s_1 + s_2) = -120 \text{ m}$$

$$a = -\frac{(F - f)}{m} = -5 \text{ m/s}^2$$

$$v^2 = u^2 + 2as = 0 + 2(-120)(-5)$$

$$v^2 = 1200 \Rightarrow v = 20\sqrt{3} = 20(1.732) = 34.64 \text{ m/s}$$

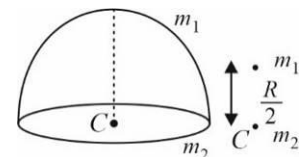
8.(0.33) Let σ be the mass per unit area (it will be same for the shell and the plate)

$$m_1 = \sigma(2\pi R^2), m_2 = \sigma(\pi R^2)$$

To find the CM, we can replace the hemispherical shell and the circular plate by point mass located at their CM as shown.

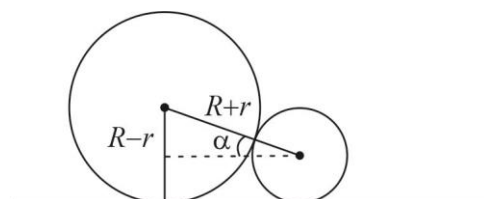
CM of the system from C

$$= \frac{m_1(R/2) + m_2(0)}{m_1 + m_2} = \frac{\sigma(2\pi R^2)(R/2)}{\sigma 3\pi R^2} = \frac{R}{3} = 0.33R$$



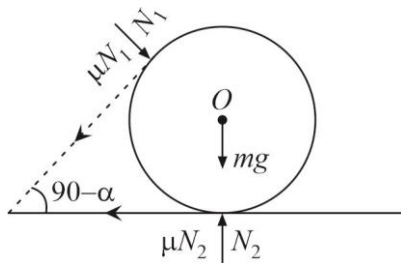
9.(0.50)

$$\sin \alpha = \frac{R - r}{R + r}$$



$$\cos \theta = \sqrt{1 - \sin^2 \alpha} = \frac{2\sqrt{Rr}}{R + r}$$

$$\text{Smaller roller : } \tau \text{ about } O \Rightarrow \mu N_1 r = \mu N_2 r$$



$$\Rightarrow N_1 = N_2$$

Forces along X : $\mu N_2 + \mu N_1 \cos (90 - \alpha) = N_1 \cos \alpha$

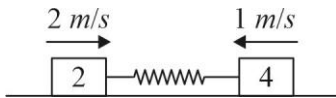
$$\mu N_1 + \mu N_1 \sin \alpha = N_1 \cos \alpha$$

$$\mu \left(1 + \frac{R-r}{R+r} \right) = \frac{2\sqrt{Rr}}{R+r} \Rightarrow \mu = \sqrt{\frac{r}{R}} = \frac{1}{2} = 0.5$$

10.(49.5)

$$V_{CM} = \frac{2 \times 6 + 3 \times 4}{2 + 4} = \frac{24}{6} = 4 \text{ m/s}$$

In the figure of C.M at $t = 0$



$$\text{Block perform SHM about CM with time period } T = 2\pi\sqrt{\frac{4}{3k}} = 2\pi\sqrt{\frac{4}{3000}} = \frac{4\pi}{30\sqrt{10}}$$

$$\text{At } t = \frac{T}{6} = \frac{\pi}{45\sqrt{10}}, V = \frac{V_{\max}}{2}$$

$$\text{K.E.} = \frac{1}{2} \times 2(1+4)^2 + \frac{1}{2} \times 4 \times \left(-\frac{1}{2} + 4 \right)^2 = 25 + 2 \times \frac{49}{4} = 25 + 24.5 = 49.5 \text{ J}$$

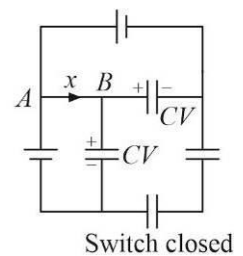
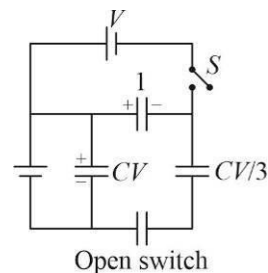
11.(3.60)

After closing the switch capacitor between B and S changes its potential from $V/3$ to V where as other capacitor connected to point B continued to maintain its voltage at V .

Let the charge flown from A to B be x then.

x = change in charge on capacitor '1'

$$= CV - \frac{CV}{3} = \frac{2CV}{3} = 3.60$$



12.(3.14)

F = thrust ; W = weight

$$kx_0 = W - F$$

If displaced by x total energy

$$E = -(W - F)x + \frac{1}{2}k(x + x_0)^2 + \frac{1}{2}mv^2 + \frac{1}{4}\rho Vv^2$$

$$\frac{dE}{dt} = 0$$

$$\Rightarrow O = (W - F) \frac{dx}{dt} + k(x + x_0) \frac{dx}{dt} + mv \frac{dv}{dt} + \frac{1}{2} \rho V v \frac{dv}{dt}$$

$$\Rightarrow a = -\frac{k}{\frac{1}{2} \rho V + m} \cdot x$$

$$T = 2\pi \sqrt{\frac{m + \frac{1}{2} \rho V}{k}} = 2\pi \sqrt{\frac{20 + \frac{1}{2} (1000)(0.16)}{400}} = \frac{2\pi}{2} = \pi$$

13.(1.78)

When mirror is absent $I = \frac{P}{4\pi a^2}$

With mirror present $I' = \left(\sqrt{I} + \frac{\sqrt{I}}{3} \right)^2 = \frac{16}{9} I \Rightarrow \frac{I'}{I} = \frac{16}{9} = 1.78$

14.(2.78)

Increase in kinetic energy of particle $= \frac{1}{2} m(2v)^2 - \frac{1}{2} mv^2 = \frac{3}{2} mv^2$

Work done by the uniform electric field, E , in going from P to $Q = (qE) \times 2a = 2qEa$

Hence, $2qEa = \frac{3}{2} mv_1^2$

or $E = \frac{3mv^2}{4qa}$

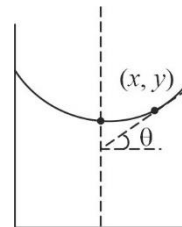
$$\frac{3}{4} \times \frac{6.6 \times 10^{-27}}{1.6 \times 10^{-19}} \times \frac{(3 \times 10^4)^2}{10} = 2.78 \text{ V/m}$$

15.(B) 16.(D)

$$\tan \theta = \frac{a_x}{a_y} = \frac{\omega_0^2 x}{g} = \frac{dy}{dx}$$

$$\frac{d^2 y}{dx^2} = \frac{\omega_0^2}{g}$$

$$\rho = \text{Radius of curvature} = \frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}{\frac{d^2 y}{dx^2}} = \frac{\left[1 + \left(\frac{\omega_0^2 x}{g} \right)^2 \right]^{3/2}}{\frac{\omega_0^2}{g}}$$



Using mirror formulae :

$$\frac{-2}{\rho_{x=0}} = \frac{1}{y_i} - \frac{1}{y_0}$$

$$\frac{-2\omega_0^2}{g} = \frac{1}{y_i} - \frac{1}{y_0} \Rightarrow y_i = \frac{gy_0}{g - 2\omega_0^2 y_0}$$

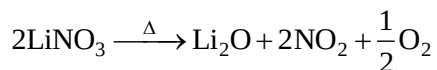
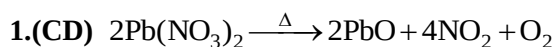
For real image $y_i < 0 \Rightarrow y_0 > \frac{g}{2\omega_0^2}$

$$17.(A) \quad \Delta q = \frac{\Delta \phi}{R} = \left[\frac{\mu_0 i}{2\pi l} \times 2\pi l \times n \right] S \times \frac{1}{R} \Rightarrow Q = \frac{\mu_0 n i S}{R}$$

$$18.(A) \quad \phi = \int (\vec{B}_1 + \vec{B}_2) \cdot \vec{dA} n = \int (\vec{B}_1 + \vec{B}_2) \cdot \vec{dl} n dA = \int \vec{B}_1 \cdot \vec{dl} n dA = \text{Depends on } i_1.$$

$$\text{As } \oint \vec{B}_2 \cdot \vec{dl} = 0 \text{ (Ampere's law)}$$

CHEMISTRY



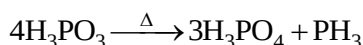
2.(ACD)

As the M – C π bonding increases the C – O bond length increases

As the electron density on the central metal atom increases, the C – O bond length increases

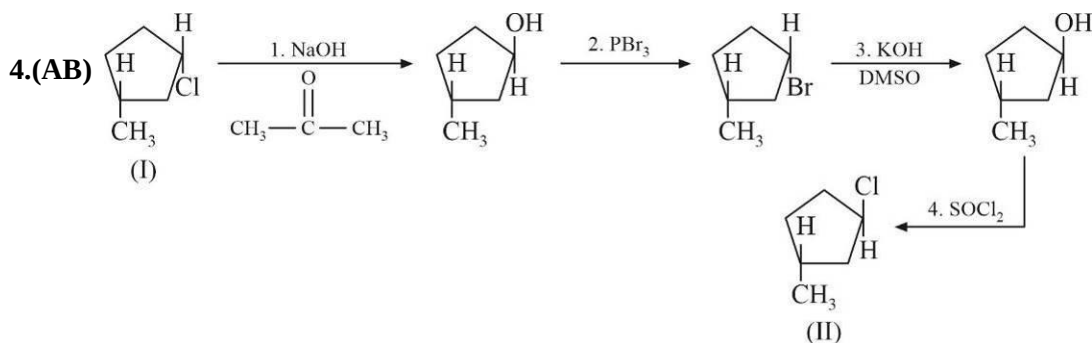
In $\text{Cr}(\text{CO})_6$; EAN = 24 + 12 = 36

3.(BC) Nitrogen being second period element, d-orbitals are not available for bonding

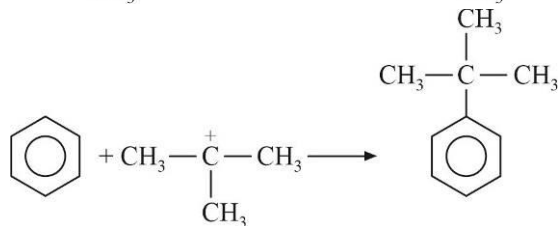
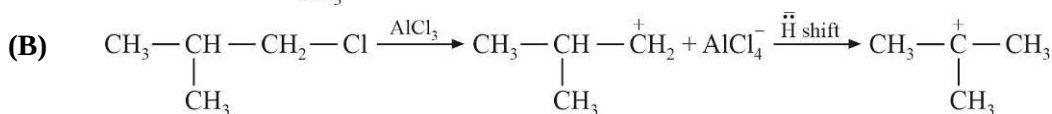
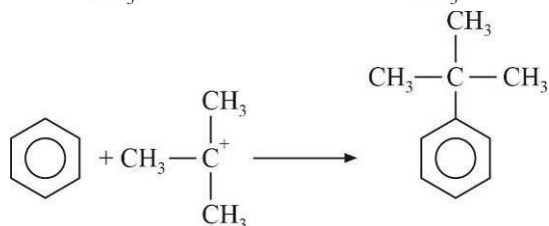
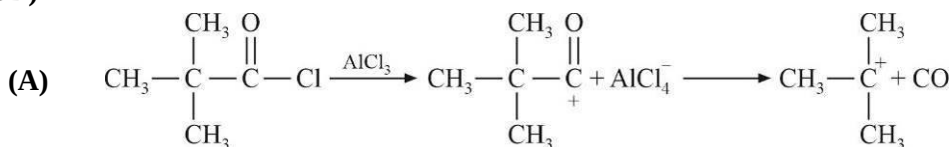


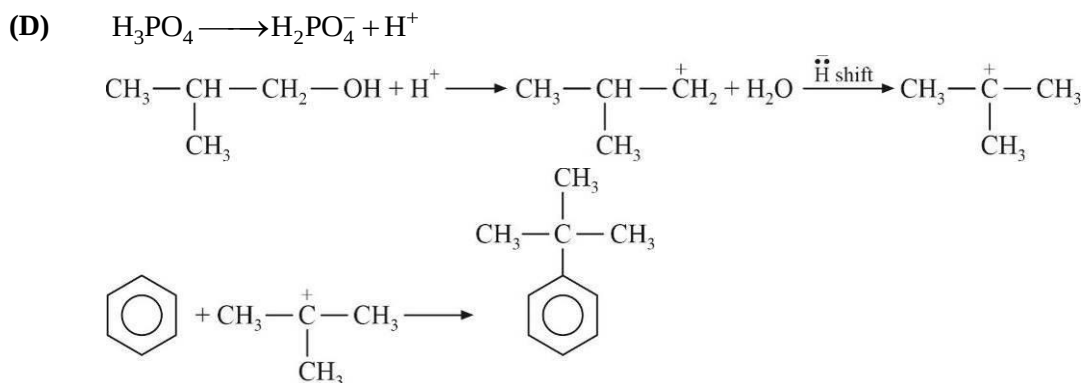
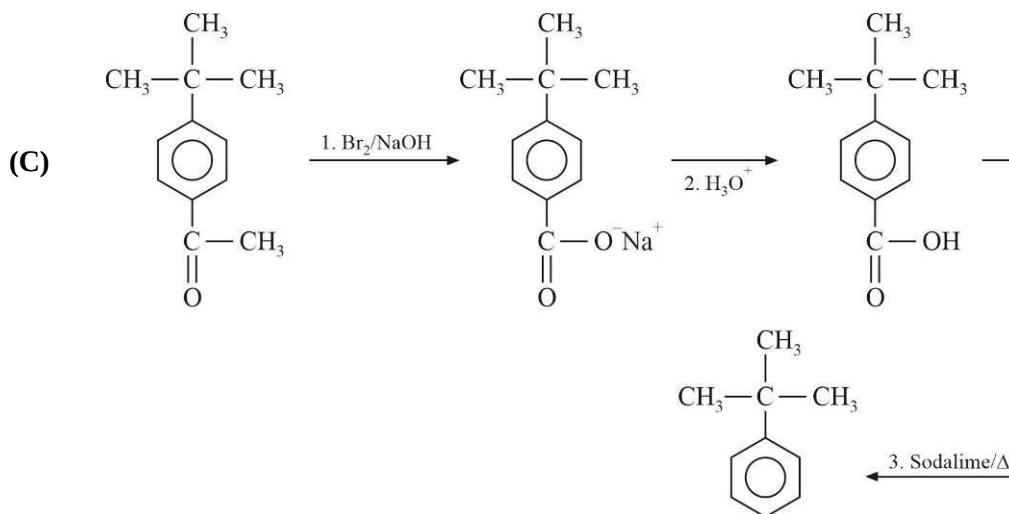
The reducing nature of hydrides increases down the group

The basic nature of oxides increases down the group.



5.(ABCD)





6.(BD) At point a,

$$P_0 V_0 = 1 \times R T_0 \Rightarrow T_0 = \frac{P_0 V_0}{R}$$

At point b,

$$P_0 \times 4V_0 = 1 \times R \times T_b \Rightarrow T_b = \frac{4P_0 V_0}{R} = 4T_0$$

At point c,

$$2P_0 \times 4V_0 = 1 \times R \times T_c \Rightarrow T_c = \frac{8P_0 V_0}{R} = 8T_0$$

$$\Delta U_{a \rightarrow b} = 1 \times \frac{3}{2} R \times (4T_0 - T_0) = \frac{9}{2} R T_0$$

$$\Delta U_{b \rightarrow c} = 1 \times \frac{3}{2} R \times (8T_0 - 4T_0) = 6 R T_0$$

$$\Delta U_{a \rightarrow c} = \Delta U_{a \rightarrow b} + \Delta U_{b \rightarrow c} = \frac{9}{2} R T_0 + 6 R T_0$$

$$\Delta U_{a \rightarrow c} = \frac{21}{2} R T_0$$

$$W_{a \rightarrow b} = -P_0 \times (4V_0 - V_0) = -3P_0 V_0 = -3R T_0$$

$$\Delta U_{a \rightarrow b} = q_{a \rightarrow b} + W_{a \rightarrow b}$$

$$\frac{9}{2} R T_0 = q_{a \rightarrow b} - 3R T_0$$

$$q_{a \rightarrow b} = \frac{9}{2}RT_0 + 3RT_0 = \frac{15}{2}RT_0$$

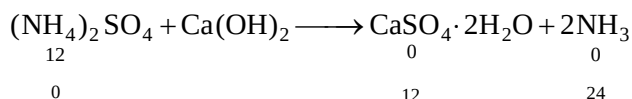
Process $a \rightarrow b$ is isobaric

$$\text{Hence } q_{a \rightarrow b} = \Delta H_{a \rightarrow b} \Rightarrow \Delta H_{a \rightarrow b} = \frac{15}{2}RT_0$$

7.(5) Paramagnetic species $\rightarrow O_2, O_2[AsF_6], KO_2, [Cu(NH_3)_4]^{2+}$ and $(NH_4)_2[CoCl_4]$

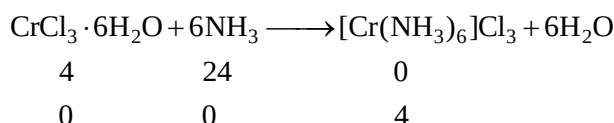
8.(3106)

$$\text{Number of moles of } (NH_4)_2SO_4 = \frac{1584}{132} = 12$$



Moles of NH_3 formed = 24

$$\text{Number of moles of } CrCl_3 \cdot 6H_2O = \frac{1066}{266.5} = 4$$



Mass of gypsum = $12 \times 172 = 2064$ gram

Mass of $[Cr(NH_3)_6]Cl_3 = 4 \times 260.5 = 1042$ gram

Total mass of gypsum and $[Cr(NH_3)_6]Cl_3 = 2064 + 1042 = 3106$ gram

9.(155)

$$d = \frac{Z \times M}{a^3 \times N_A}$$

$$Z = 4 \times \frac{99.9}{100} = 3.996$$

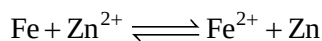
$$d = \frac{3.996 \times 40}{(0.556 \times 10^{-7})^3 \times 6 \times 10^{23}}$$

$$d = 1.55 \text{ gram/cm}^3$$

Hence $100d = 155$

10.(100)

$$E^\circ = \frac{0.06}{n} \log K$$



$$\begin{array}{cc} 1 & 0 \end{array}$$

$$\begin{array}{cc} 1-y & y \end{array}$$

$$E^\circ = -0.33 = \frac{0.06}{2} \log \frac{y}{(1-y)} \quad (1-y \sim 1)$$

$$\Rightarrow y = 1 \times 10^{-11} \Rightarrow [Fe^{2+}] = 1 \times 10^{-11} = 100 \times 10^{-13} = x \times 10^{-13} \Rightarrow x = 100$$

11.(150)

$$\Delta U = q + W$$

$$W = \Delta U - q$$

$$= 400 - 5059.8$$

$$W = -4659.8 \text{ J}$$

$$W = -\int P dV = -\int_1^{V_2} \frac{20}{V} dV$$

$$W = -20 \ln V_2$$

$$W = -20 \times 2.3 \log V_2 \text{ litre-atm}$$

$$W = -20 \times 2.3 \times 101.3 \log V_2 \text{ Joule}$$

$$\Rightarrow -4659.8 = -20 \times 2.3 \times 101.3 \log V_2$$

$$\Rightarrow \log V_2 = 1 \quad \Rightarrow V_2 = 10 \text{ litre}$$

$$\text{Hence, } 15V_2 = 150$$

12.(50) $P_s = x_A P_A^\circ + x_B P_B^\circ$

$$109.13 = x_A \times 200 + (1 - x_A) \times 75$$

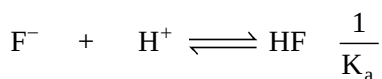
$$\Rightarrow x_A = 0.273$$

$$y_A = \frac{x_A \times P_A^\circ}{P_s}$$

$$y_A = \frac{0.273 \times 200}{109.13}$$

$$y_A = 0.5$$

$$\text{Mole percent of A} = y_A \times 100 = 0.5 \times 100 = 50\%$$

13.(50) $\text{MgF}_{2(s)} \rightleftharpoons \underset{S}{\text{Mg}}^{2+} + \underset{2S}{2\text{F}^-}, K_{sp}$ 

$$2S \quad 10^{-4}$$

$$\left(\frac{K_{sp}}{S} \right)^{1/2} \quad 10^{-4} \quad 2S$$

$$\frac{1}{K_a} = \frac{2s}{10^{-4} \times \left(\frac{K_{sp}}{S} \right)^{1/2}}$$

$$\frac{1}{3.5} \times 10^4 = \frac{2s^{3/2}}{10^{-4} \times (K_{sp})^{1/2}}$$

$$S^{3/2} = \frac{1}{7} \times (K_{sp})^{1/2}$$

$$S^3 = \frac{1}{49} \times K_{sp}$$

$$S^3 = \frac{1}{49} \times 6.125 \times 10^{-9}$$

$$S = 5 \times 10^{-4}$$

$$S = 50 \times 10^{-5} \text{ mol/litre} = Y \times 10^{-5}$$

$$\Rightarrow \text{Hence } Y = 50$$

14.(30) For solute S

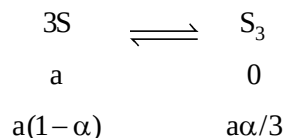
$$\Delta T_b = i \times 0.1 \times K_b$$

For other solute

$$\Delta T_b = 1 \times 0.08 \times K_b$$

$$\text{Hence, } i \times 0.1 \times K_b = 0.08 \times K_b$$

$$\Rightarrow i = 0.8$$



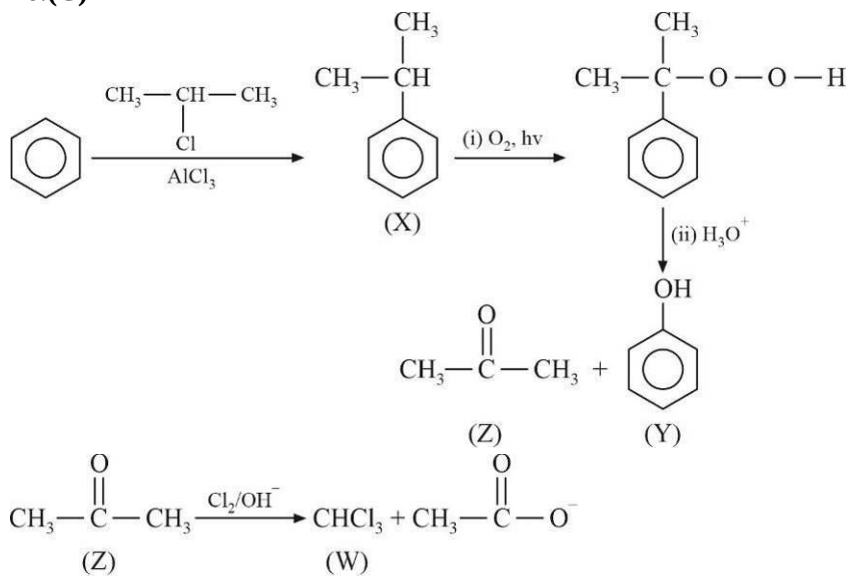
$$i = \frac{a(1-\alpha) + \frac{a\alpha}{3}}{a}$$

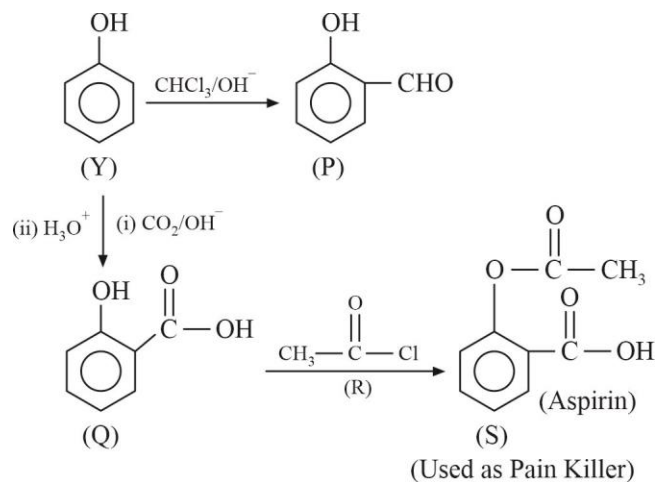
$$i = 1 - \frac{2\alpha}{3}$$

$$0.8 = 1 - \frac{2\alpha}{3}$$

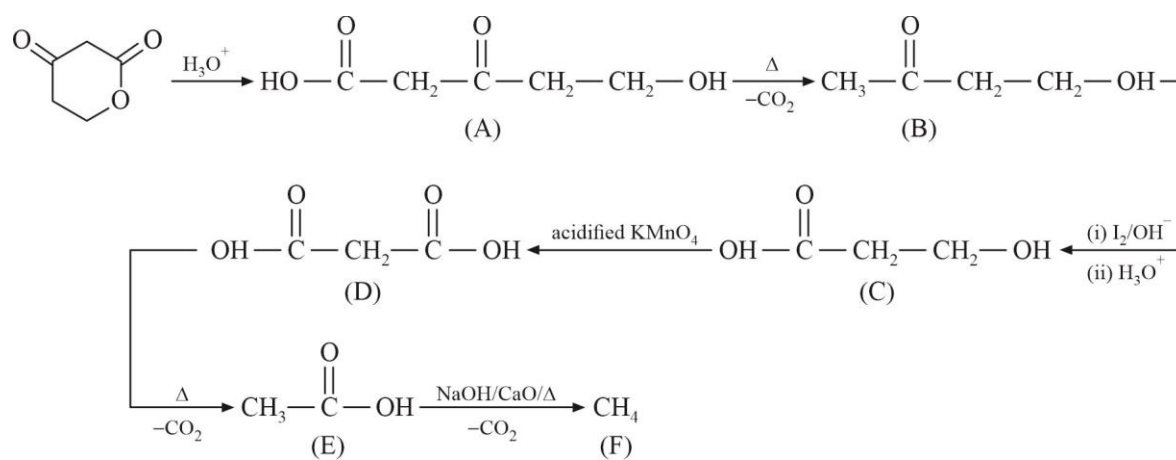
$$\frac{2\alpha}{3} = 0.2 \Rightarrow \alpha = 0.3 = 30 \times 10^{-2} \Rightarrow Y = 30$$

15.(A) 16.(C)





17.(D) 18.(B)



MATHEMATICS

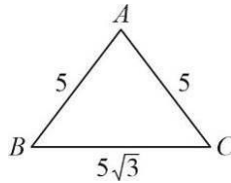
1.(ABD)

2.(AB) Given that $a = 5\sqrt{3}$, $b = 5$, $c = 5$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{5^2 + 5^2 - (5\sqrt{3})^2}{2 \times 5 \times 5}$$

$$\Rightarrow \cos A = -\frac{25}{50} = -\frac{1}{2}$$

$$A = 120^\circ$$



$$\text{Area of } \triangle ABC = \frac{1}{2}bc \sin A = \frac{1}{2} \times 5 \times 5 \times \sin 120^\circ = \frac{1}{2} \times 5 \times 5 \times \frac{\sqrt{3}}{2} = \frac{25\sqrt{3}}{4}$$

$$\text{Radius of incircle of } \triangle ABC = r = \frac{\Delta}{s} = \frac{\frac{25\sqrt{3}}{4}}{\frac{10 + 5\sqrt{3}}{2}} = \frac{5}{2}(2\sqrt{3} - 3)$$

$$\text{Radius of circumcircle of } \triangle ABC = R = \frac{abc}{4\Delta} = \frac{5\sqrt{3} \times 5 \times 5}{4 \times \frac{25\sqrt{3}}{4}} = 5 \text{ cm}$$

3.(BCD)

$$(A) \quad \vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix} = -\hat{i} + 3\hat{j} + 5\hat{k}$$

Equation of plane is

$$-1(x - 3) + 3(y - 2) + 5(z + 1) = 0$$

$$-x + 3y + 5z + 2 = 0$$

$$x - 3y - 5z = 2$$

$$(B) \quad \cos \theta = \frac{|2 - 2 - 1|}{\sqrt{6} \sqrt{6}} = \frac{1}{6}$$

(D) Any plane passing through the intersection of P_1 and P_2 is

$$(2x - y + z - 2) + \lambda(x + 2y - z - 3) = 0$$

$$(1, 3, -2) \text{ lies on it, therefore } \lambda = \frac{5}{6}$$

Equation of required plane is

$$(2x - y + z - 2) + \frac{5}{6}(x + 2y - z - 3) = 0 \Rightarrow 17x + 4y + z - 27 = 0$$

4.(ABC)

$$(A) \quad \lim_{x \rightarrow 0} f(x^2) = \lim_{x \rightarrow 0} (f(x))^2 \Rightarrow f(0) = 1$$

Continuity forces $f(x) = 1$ for all x

$$(B) \quad \text{If } f(a) = 0 \Rightarrow f(a^{2^{-n}}) = (f(a))^{2^{-n}} \Rightarrow f(1) = 0$$

Assume for some b $f(b) > 0$

$$\Rightarrow f(b^{2^{-n}}) = (f(b))^{2^{-n}} \Rightarrow f(1) = 1 \text{ contradiction}$$

(C) Define $\forall t \in (2, \infty)$

$$f_t(x) = \begin{cases} x^{t-1} & x \in (0, 1) \\ 1 & x = 1 \\ x^{-(t+1)} & x \in (1, \infty) \end{cases}$$

$f_t(x)$ satisfies the given property that $f_t(x^2) = (f_t(x))^2 \forall x \in (0, \infty)$

$$\int_0^{\infty} f_t(x) dx = \frac{1}{t} + \frac{1}{t} = \frac{2}{t} < 1$$

Hence, infinite such functions

5.(ABCD)

$$f(f(x)) = x \Rightarrow f(f(0)) = 0 \Rightarrow f(1) = 0$$

$$\text{Now } f'(f(x)) \cdot f'(x) = 1 \Rightarrow f \text{ is decreasing and } f(1) = 0$$

$f : [0, 1] \rightarrow [0, 1]$ is invertible

$$\text{Let } I = \int_0^1 (x - f(x))^{2020} dx, \text{ put } f(x) = t \Rightarrow I = \frac{1}{2021}$$

6.(BCD)

$$\text{Notice } f(0) = 0 \text{ and } f'(x) = 2f(x) \Rightarrow f(x) = 0 \forall x$$

$$7.(1) (\log_3 7)^{8 \log(\log_9 49)^7} \times 5^{-8 \log_5 7}$$

$$(\log_3 7)^{8 \log(\log_3 7)^7} \times 5^{\log_5 7^{-8}}$$

$$(\log_3 7)^{\log(\log_3 7)^{7^8}} \times 7^{-8} = 7^8 \times 7^{-8} = 1$$

8.(216) Sum of all digits in set is $0 + 1 + 2 + 3 + 4 + 5 = 15$

Case I

If number is formed using 1, 2, 3, 4, 5 then number of numbers = $5! = 120$

Case II

If number is formed using 0, 1, 2, 4, 5 then number of numbers = $4 \times 4! = 96$

So, total numbers = $120 + 96 = 216$

9.(3638)

Common elements in X and Y are 18, 38, 58, 78,, 8078

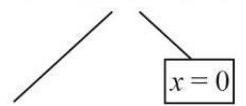
Number of common elements in X and Y is 404

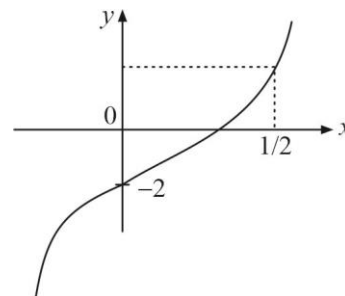
So, Number of elements in XUY is

$$2021 + 2021 - 404 = 3638$$

$$10.(1) \cos^{-1} \left(\frac{-x/2}{1 + \frac{x}{2}} + \frac{x}{1+x} \right) = \cos^{-1} \left(\frac{x^2}{1-x} - \frac{x \times \frac{x}{2}}{1 - \frac{x}{2}} \right)$$

$$\Rightarrow \frac{-x}{2+x} + \frac{x}{1+x} = \frac{x^2}{1-x} - \frac{x^2}{2-x}$$





$$\Rightarrow \frac{1}{1+x} - \frac{1}{2+x} = \frac{x}{1-x} - \frac{x}{2-x}$$

$$\Rightarrow \frac{x^3 + 2x^2 + 5x - 2}{y} = 0 \Rightarrow y' = \frac{3x^2 + 4x + 5}{a > 0 \text{ and } D < 0}$$

$$\Downarrow$$

$$y' > 0$$

and $y\left(\frac{1}{2}\right) = +ve$

$\Downarrow$

There exist a unique solution in $\left(0, \frac{1}{2}\right)$

11.(5) $P_n = \left(\left(\frac{n+2}{n}\right)\left(\frac{n+4}{n}\right)\left(\frac{n+6}{n}\right)\dots\left(\frac{n+2n}{n}\right)\right)^{\frac{1}{2}}$

$$\Rightarrow P_n = e^{\frac{1}{n} \sum_{r=1}^n \ln\left(1 + \frac{2r}{n}\right)} \Rightarrow L = e^{\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \ln\left(1 + \frac{2r}{n}\right)}$$

$$\Rightarrow L = e^{\int_0^1 \ln(1+2x) dx} \Rightarrow I = \int_0^1 \frac{1}{1} \cdot \frac{\ln(1+2x)}{1} dx$$

$$I = \ln(1+2x) \times x \Big|_0^1 - \int_0^1 \frac{1 \times 2x}{2x+1} dx \Rightarrow I = \frac{3}{2} \ln 3 - 1$$

$$\therefore L = e^{\frac{3}{2} \ln 3 - 1} \Rightarrow L = \frac{\sqrt{27}}{e}$$

$$\Rightarrow Le = \sqrt{27}$$

$$\Downarrow$$

$$[Le] = 5$$

12.(5) $\vec{a} \cdot \vec{c} = |\vec{a}| |\vec{c}| \cos \beta$

$$\vec{a} \cdot \vec{c} = 3 \cos \beta$$

$$\vec{b} \cdot \vec{c} = 3 \cos \beta$$

$$\Rightarrow \vec{c} = m \vec{a} + n \vec{b} + \vec{a} \times \vec{b}$$

$$\vec{a} \cdot \vec{c} = m = 3 \cos \beta$$

$$\vec{b} \cdot \vec{c} = n = 3 \cos \beta$$

$$\Rightarrow \text{Let } m = n = 3 \cos \beta = \lambda \Rightarrow \vec{c} = \lambda \vec{a} + \lambda \vec{b} + \vec{a} \times \vec{b}$$

$$\Rightarrow |\vec{c}|^2 = \lambda^2 + \lambda^2 + |\vec{a} \times \vec{b}|^2 = 9 \Rightarrow 2\lambda^2 + 1 = 9$$

$$\Rightarrow \lambda^2 = 4$$

$$\Downarrow$$

$$9 \cos^2 \beta = 4$$

13.(2) $\sqrt{3}p \sin x + 2q \cos x = r$

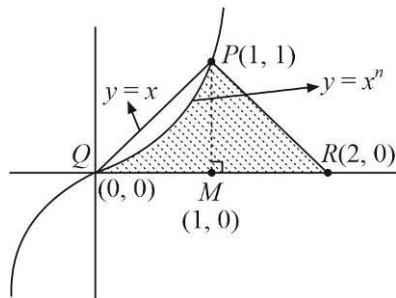
$\Rightarrow \sqrt{3}p \sin \alpha + 2q \cos \alpha = r$

$\sqrt{3}p \sin \beta + 2q \cos \beta = r$

$\Rightarrow \sqrt{3}p(\sin \alpha - \sin \beta) + 2q(\cos \alpha - \cos \beta) = 0 \quad \Rightarrow \quad \sqrt{3}p \cos \frac{\alpha + \beta}{2} = 2q \sin \frac{\alpha + \beta}{2}$

$\Rightarrow \frac{p}{q} = \frac{2}{\sqrt{3}} \tan \frac{\alpha + \beta}{2} \quad \Rightarrow \quad \frac{p}{q} = \frac{2}{\sqrt{3}} \times \frac{1}{\sqrt{3}}$

14.(9)



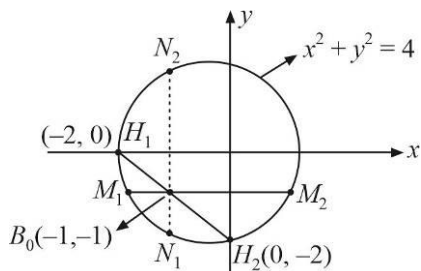
$\Rightarrow \triangle PQR \text{ is right angled at } P \quad \Rightarrow \quad A_1 = \frac{1}{2} \times PQ \times PR = \frac{1}{2} \times \sqrt{2} \times \sqrt{2} = 1$

$\Rightarrow \text{Area of shaded portion} = 0.6 \times A_1$

$\Rightarrow \int_0^1 x^n dx + \frac{1}{2} \times 1 \times 1 = 0.6$

$\Rightarrow \frac{1}{n+1} = 0.6 - 0.5 \quad \Rightarrow \quad \frac{1}{n+1} = \frac{1}{10} \quad \Rightarrow \quad n = 9$

15.(C)



Equation of chord of contact to a circle drawn from (h, k) is:

$hx + ky = 4$

(1) Equation of $M_1M_2 : y = -1 \quad \Rightarrow \quad M_3(0, -4)$

(2) Equation of $N_1N_2 : x = -1 \quad \Rightarrow \quad N_3(-4, 0)$

(3) Equation of $H_1H_2 : y + 1 = -(x + 1)$
 $x + y = -2 \quad \Rightarrow \quad H_3(-2, -2)$

16.(D) Tangent at 'P' is given by

$$x \cos \theta + y \sin \theta = 2$$

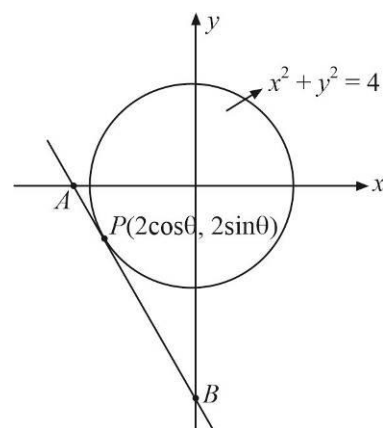
$$\Rightarrow A\left(\frac{2}{\cos \theta}, 0\right), B\left(0, \frac{2}{\sin \theta}\right)$$

$\Rightarrow$ Let 'M' is midpoint of AB and let M be (h, k)

$$\Rightarrow h = \frac{\frac{2}{\cos \theta} + 0}{2}, k = \frac{0 + \frac{2}{\sin \theta}}{2}$$

$$\Rightarrow \cos \theta = \frac{1}{h}, \sin \theta = \frac{1}{k}$$

$$\Rightarrow \cos^2 \theta + \sin^2 \theta = \frac{1}{h^2} + \frac{1}{k^2} \Rightarrow x^2 + y^2 = x^2 y^2$$



17.(A) Probability = $\frac{\text{Number of favourable arrangements}}{\text{Total number of arrangements}} = \frac{D_3}{4!}$

$\rightarrow$ De-arranging of 3 objects

$$D_3 = 3! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!}\right) \Rightarrow \frac{2}{24} = \frac{1}{12}$$

$$D_3 = 2$$

18.(A) Number of total arrangements = 4!

Number of arrangements favouring $E_2 \cap E_3 \cap E_4$:

$\Rightarrow$ constraints of P_1 and P_4 are same

$\Rightarrow \underline{P_1} \underline{P_3} _ _ \Rightarrow$ No such arrangement is possible

$\underline{P_1} \underline{P_4} \underline{P_2} \underline{P_3} \Rightarrow$ No such that arrangement is possible

There are

$\Rightarrow$ No such arrangement where P_1, P_4

Occupies 1st position

$$\therefore \text{Probability } (E_2 \cap E_3 \cap E_4) = \frac{2}{4!} = \frac{1}{12}$$

Constraints of P_2 and P_3 are same

$\rightarrow \underline{P_2} \underline{P_4} \underline{P_1} \underline{P_3} \checkmark$

$\rightarrow \underline{P_3} \underline{P_1} \underline{P_4} \underline{P_2} \checkmark$